ULTIMATE AWS CLOUD PRACTITIONER GOD-MODE CHEAT SHEET (FULL SYLLABUS, MAX DETAIL)

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1. CORE AWS SERVICES

Compute

- **EC2 (Elastic Compute Cloud)**
  - Virtual server with full OS control. Supports Windows/Linux.
  - Pricing: On-Demand, Reserved, Spot, Dedicated Host/Instance.
  - Use cases: legacy applications, lift-and-shift, batch processing, high-performance compute.
  - Features: Auto Scaling, Load Balancing, Placement Groups, EC2 Image Builder.
  - Key comparison: EC2 vs Lambda – EC2 is persistent and configurable; Lambda is ephemeral and event-driven.
- **Lambda**
  - Serverless. Run code in response to triggers (API Gateway, S3, EventBridge).
  - 1M free requests/month. Max 15 minutes per execution.
  - Scales automatically. Use cases: image processing, CRON jobs, lightweight APIs.
  - Key Limits: 10GB memory, 6MB input/output payload (synchronous), 250MB deployment package.
- **Elastic Beanstalk**
  - PaaS for developers. Deploy code in Node.js, Java, Python, etc.
  - Manages provisioning, scaling, health monitoring.
  - Good for fast deployment without deep infrastructure knowledge.
- **Lightsail**
  - Simplified EC2 with preconfigured templates. Includes networking, static IP, firewall.
  - Good for blogs, ecommerce, test environments, and simple VPS needs.

Storage

- **S3 (Simple Storage Service)**
    - Object-based storage. 11 9's durability. Eventual consistency.
    - Use for static website hosting, backups, file distribution.
    - Storage Classes: Standard, Intelligent-Tiering, IA, One Zone-IA, Glacier, Glacier Deep Archive.
    - Encryption: SSE-S3, SSE-KMS, Client-side.
    - Features: Lifecycle rules, versioning, cross-region replication (CRR), same-region replication (SRR), Object Lock, MFA delete.
- **EBS (Elastic Block Store)**
    - Block storage for EC2. Persistent and high performance.
    - Volumes: gp3 (general purpose), io2 (high IOPS), st1 (throughput), sc1 (cold storage).
    - Features: Snapshots, encryption, resizing, multi-attach for io2.
- **EFS (Elastic File System)**
    - Shared file storage for Linux EC2.
    - Highly available across AZs. Elastic and scalable.
    - Supports burst throughput or provisioned throughput modes.

Database

- **RDS (Relational Database Service)**
    - Managed SQL databases: MySQL, PostgreSQL, Oracle, SQL Server, MariaDB.
    - Features: Automated backups (7–35 days), snapshots, replication, Multi-AZ (HA), Read Replicas (for read scaling).
- **Aurora**
    - High-performance managed DB. Compatible with MySQL/PostgreSQL.
    - 5x faster than MySQL on RDS. Auto-healing, auto-scaling, fault-tolerant.
    - Aurora Serverless: Pay-per-query option. Scales to zero.
    - Global Database: multi-region deployment.
- **DynamoDB**

    - Serverless NoSQL key-value store. 99.999% availability.
    - Features: On-Demand/Provisioned capacity, Global Tables, TTL, Streams, DAX (in-memory caching), PartiQL.
    - Use cases: IoT, real-time bidding, gaming leaderboards.
- **Redshift**
    - OLAP warehouse. Columnar DB engine. Fast queries over petabytes.
    - Supports BI tools, Redshift Spectrum for S3 queries.
    - Features: Concurrency Scaling, RA3 instances with managed storage, Federated Query.

Networking

- **VPC (Virtual Private Cloud)**
    - Logical isolation of AWS network. Build your own DC in the cloud.
    - Components: Subnets, Route Tables, Internet/NAT Gateways, NACLs (stateless), Security Groups (stateful).
    - Peering and Transit Gateway: for VPC-to-VPC communication.
- **Subnets**
    - Public: internet-facing (via IGW). Private: internal, secure (via NAT).
    - Best practice: isolate tiers (Web, App, DB) in separate subnets.
- **Route 53**
    - Domain registration, DNS, traffic routing.
    - Policies: Simple, Latency-based, Weighted, Failover, Geolocation, Multivalue Answer.
    - Health checks and DNS failover support.
- **CloudFront**
    - Global CDN. Works with S3, ALB. Reduces latency via Edge locations.
    - Supports geo-restrictions, signed URLs/cookies, Lambda@Edge.
- **ELB (Elastic Load Balancing)**
    - ALB (HTTP/HTTPS), NLB (TCP/UDP), CLB (legacy).
    - ALB supports routing based on path/host; NLB supports static IPs, high TPS.

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4. BILLING & COST MANAGEMENT

Pricing Models

- **On-Demand**: Pay per second. Best for spiky workloads or unpredictable usage.
- **Reserved Instances**: 1-3 year commitment. Up to 75% discount. Types: Standard, Convertible, Scheduled.
- **Spot Instances**: Unused EC2. Up to 90% discount. Can be interrupted with 2-min warning.
- **Savings Plans**: Flexible pricing model. Compute SP (EC2, Lambda, Fargate), EC2 Instance SP (specific types).

Free Tier

- 12-month: EC2 750 hrs/mo (t2.micro/t3.micro), S3 5GB, RDS 750 hrs/mo.
- Always free: Lambda 1M requests/month, DynamoDB 25GB storage, CloudWatch metrics.

Cost Tools

- **Pricing Calculator**: Estimate future costs. Use scenarios and exportable reports.
- **Cost Explorer**: Visualize usage trends, create usage reports, forecast spend.
- **Budgets**: Set budget alerts for cost, usage, RI/SP coverage.
- **Billing Dashboard**: View linked account charges. Filter by service, region.
- **Cost Categories**: Group usage into business units/projects.
- **Tags**: Key-value metadata. Use for cost allocation and automation.

Consolidated Billing

- Group accounts under AWS Organizations.
- Share RI and SP benefits. Enable centralized billing.
- Separate security/account controls but combine charges.

Support Plans

Plan	Features
Basic	Free, no SLA, access to docs/forums
Developer	$29/mo+, email support, best-effort guidance
Business	$100/mo+, 24x7 support, Trusted Advisor, AWS Health
Enterprise	$15k/mo+, TAM, Concierge, infrastructure event management

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5. SECURITY, MONITORING & GOVERNANCE (EXPANDED)

IAM (Identity & Access Management)

- Users: Permanent credentials (username/password or access keys).
- Groups: Organize users with similar roles.
- Roles: Assign temporary credentials to users/services.
- Policies: JSON documents (inline or managed).
- Principle of Least Privilege: Always grant minimum required permissions.
- IAM Access Analyzer: Detect unintended access to your resources.
- Identity Federation: Use corporate credentials (SAML, Cognito, AD).

Security Services

- **KMS (Key Management Service)**: Create/manage encryption keys.
- **CloudHSM**: Dedicated hardware module for key management.
- **Secrets Manager**: Rotate, manage, retrieve DB credentials, API keys.
- **Parameter Store**: Store config data/secrets. Free tier available.

Monitoring & Audit

- **CloudTrail**: Logs all API activity. Stores in S3, supports event filters.
- **CloudWatch**: Monitors logs, metrics, events, alarms. Central to automation and visibility.
- **AWS Config**: Tracks historical changes to AWS resources.
- **AWS Audit Manager**: Automate evidence collection for audits.

Protection & Compliance

- **AWS Shield**: DDoS protection. Standard (free), Advanced (with SLAs).
- **WAF**: Protect web apps from SQLi, XSS, bad bots.
- **GuardDuty**: Detect threats using machine learning.
- **Macie**: Scans S3 for PII (names, SSNs, etc.).
- **Inspector**: Automatically assesses EC2/containers for vulnerabilities.
- **Security Hub**: Central view for findings from GuardDuty, Inspector, Macie, etc.

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6. AWS GLOBAL INFRASTRUCTURE (REVIEWED)

- Regions: Isolated locations (e.g., us-east-1, ap-south-1). Choose based on latency/compliance.
- Availability Zones (AZ): Redundant DCs within a region. Use for HA.
- Edge Locations: Content delivery via CloudFront.
- Local Zones: Edge compute near cities for low-latency apps.
- Wavelength: For 5G use cases (real-time gaming, AR).
- Outposts: AWS infra in your on-prem DC.
- Snow Family: Edge computing + data transfer appliances (Snowcone, Snowball, Snowmobile).

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7. WELL-ARCHITECTED FRAMEWORK (DETAILED)

1. Operational Excellence

- Define & automate processes.
- Monitor systems using CloudWatch/CloudTrail.
- Improve via game days/post-mortems.
- Use Infrastructure-as-Code (IaC) via CloudFormation.

2. Security

- Implement identity/permissions with IAM.
- Enable encryption at-rest/in-transit.
- Monitor with GuardDuty, CloudTrail.

- Apply patches regularly using Systems Manager.

3. Reliability

- Deploy across AZs/Regions.
- Use load balancing, Route 53 failover.
- Perform backups (EBS snapshots, RDS automated).
- Use health checks, failover logic.

4. Performance Efficiency

- Choose optimal instance types and storage.
- Use caching (CloudFront, ElastiCache).
- Go serverless where possible (Lambda, Fargate).
- Measure and evolve with trusted metrics.

5. Cost Optimization

- Use spot/reserved/Savings Plans.
- Turn off idle resources (dev/test).
- Use cost tools like Explorer, Budgets.
- Right-size resources using Trusted Advisor.

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8. ADDITIONAL SERVICES AND TOOLS

- **CloudFormation**: Infrastructure as Code.
- **CDK (Cloud Development Kit)**: Define infra in TypeScript, Python, etc.
- **Control Tower**: Govern multi-account AWS environments.
- **AWS Organizations**: Central management of linked accounts.
- **Service Catalog**: Curate and manage pre-approved app stacks.
- **Systems Manager**: Patch Manager, Session Manager, State Manager.
- **AWS Backup**: Centralized backup for EBS, RDS, DynamoDB, etc.
- **Marketplace**: Buy SaaS, AMIs, containers.
- **Trusted Advisor**: 5 pillar best practices insights.
